JavaScript Comparisons and Conditionals Review

Comparisons and the `null` and `undefined` Data Types

- **Comparisons and `undefined`**: A variable is `undefined` when it has been declared but hasn't been assigned a value. It's the default value of uninitialized variables and function parameters that weren't provided an argument. `undefined` converts to `NaN` in numeric contexts, which makes all numeric comparisons with `undefined` return `false`.

```
console.log(undefined < 0); // false (NaN < 0 is false)
console.log(undefined == ""); // false (NaN >= 0 is false)
```

```
js/sandbox.8a7d01a44.js:1:182197)
    at je.evaluateModule (https://2-19-8-sandpack.codesandbox.io/static/js/sandbox.8a7d01a44.js:1:181696)
    at https://2-19-8-sandpack.codesandbox.io/static/js/sandbox.8a7d01a44.js:1:354442
    at Generator.next (<anonymous>)
false
false
false
false
false
false
```

- **Comparisons and `null`**: The `null` type represents the intentional absence of a value. `null` converts to `0` in numeric contexts, which may result in unexpected behavior in numeric comparisons:

```
console.log(null < 0); // false (0 < 0 is false)
console.log(null >= 0); // true (0 >= 0 is true)
```

```
false
true
```

- When using the equality operator (`==`), `null` and `undefined` only equal each other and themselves:

```
console.log(null == undefined); // true
console.log(null == 0); // false
console.log(undefined == NaN); // false
```

```
true
false
false
```

- However, when using the strict equality operator (`===`), which checks both value and type without performing type coercion, `null` and `undefined` are not equal:

```
console.log(null === undefined); // false
```

```
false
```

`switch` Statements

- **Definition:** A `switch` statement evaluates an expression and matches its value against a series of `case` clauses. When a match is found, the code block associated with that case is executed. A `break` statement should be placed at the end of each case, to terminate its execution and continue with the next. The `default` case is an optional case and only executes if none of the other cases match. The `default` case is placed at the end of a `switch` statement.

```
  case 1:
    console.log("It's Monday! Time to start the week strong.");
    break;
  case 2:
    console.log("It's Tuesday! Keep the momentum going.");
    break;
  case 3:
    console.log("It's Wednesday! We're halfway there.");
    break;
  case 4:
    console.log("It's Thursday! Almost the weekend.");
    break;
  case 5:
    console.log("It's Friday! The weekend is near.");
    break;
  case 6:
    console.log("It's Saturday! Enjoy your weekend.");
    break;
  case 7:
    console.log("It's Sunday! Rest and recharge.");
    break;
  default:
    console.log("Invalid day! Please enter a number between 1 and 7.");
```

```
"It's Wednesday! We're halfway there."
```

Assignment

- [x] Review the JavaScript Comparisons and Conditionals topics and concepts.